

M15 Guided Labs

Follow once; save observed evidence

OPTION B | 16 LESSON ROUTES

Live cloud claims require actual execution.

DE15-01 - Cloud fundamentals for data engineers

GUIDED | Teaching Book p6 | route page 2

Starting state

Open Teaching Book page 6. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Draw source -> durable state -> compute/orchestration -> consumer, then add identity boundary and monitoring signal.

Expected result

architecture map with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

Record what you actually observed. If cloud access is unavailable, label it OFFLINE PLAN or SIMULATOR; do not invent LIVE FABRIC evidence.

DE15-02 - Identity, IAM and least privilege

GUIDED | Teaching Book p9 | route page 3

Starting state

Open Teaching Book page 9. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Map persona -> required task -> minimum Fabric/OneLake permission. Include one permission you deliberately refuse.

Expected result

least-privilege permission matrix with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

Record what you actually observed. If cloud access is unavailable, label it OFFLINE PLAN or SIMULATOR; do not invent LIVE FABRIC evidence.

DE15-03 - Azure Storage concepts and OneLake

GUIDED | Teaching Book p12 | route page 4

Starting state

Open Teaching Book page 12. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

For three sources choose copy or OneLake shortcut; state ownership, freshness, replay and failure implications.

Expected result

copy vs shortcut decision with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

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DE15-04 - Fabric workspace and item model

GUIDED | Teaching Book p15 | route page 5

Starting state

Open Teaching Book page 15. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Name dev workspace items: Lakehouse, notebook, pipeline and serving surface. State input/output/state/owner/proof for each.

Expected result

workspace/item inventory with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

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DE15-05 - OneLake paths, files, tables and shortcuts

GUIDED | Teaching Book p18 | route page 6

Starting state

Open Teaching Book page 18. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Map Files, Tables and one shortcut. State which paths are raw, table-managed, external, and who owns lifecycle.

Expected result

OneLake path and shortcut map with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

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DE15-06 - Fabric Lakehouse and Delta tables

GUIDED | Teaching Book p21 | route page 7

Starting state

Open Teaching Book page 21. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Use the normal Retail Orders fixture. State table grain, accepted/quarantine rules and prove 7 raw -> 4 accepted + 3 quarantined; accepted amount 3300.

Expected result

Lakehouse/Delta contract with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

Record what you actually observed. If cloud access is unavailable, label it OFFLINE PLAN or SIMULATOR; do not invent LIVE FABRIC evidence.

DE15-07 - Fabric notebooks

GUIDED | Teaching Book p24 | route page 8

Starting state

Open Teaching Book page 24. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Write parameters/config -> read -> validate -> transform -> write -> reconcile. Mark side effects and clean-session rerun proof.

Expected result

rerunnable notebook flow with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

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DE15-08 - Spark in Fabric

GUIDED | Teaching Book p27 | route page 9

Starting state

Open Teaching Book page 27. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Record production/default Runtime 1.3 GA and optional compatibility tests. then list package/API dependencies and

Expected result

Fabric runtime compatibility checklist with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

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DE15-09 - Fabric Data Pipelines

GUIDED | Teaching Book p30 | route page 10

Starting state

Open Teaching Book page 30. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Design ingest -> validate -> transform -> publish with parameters, retries, blocking failure path and checkpoint rule.

Expected result

pipeline dependency and failure graph with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

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DE15-10 - SQL analytics endpoint and warehouse choice

GUIDED | Teaching Book p33 | route page 11

Starting state

Open Teaching Book page 33. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Choose SQL analytics endpoint or Warehouse for two workload needs. State write/read ownership, serving grain and validation query.

Expected result

SQL endpoint vs Warehouse choice with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

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DE15-11 - Medallion architecture in Fabric

GUIDED | Teaching Book p36 | route page 12

Starting state

Open Teaching Book page 36. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Map Bronze raw -> Silver validated -> Gold serving with grain, bad-row handling, rerun and reconciliation at each boundary.

Expected result

medallion architecture with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

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DE15-12 - Incremental loads and MERGE in cloud

GUIDED | Teaching Book p39 | route page 13

Starting state

Open Teaching Book page 39. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Run the normal simulator: base accepted state 4 rows/3300; update O2 + insert O8 -> 5 rows/4250; exact rerun no-op. State failure-before-publish checkpoint rule.

Expected result

incremental MERGE state transition with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

Record what you actually observed. If cloud access is unavailable, label it OFFLINE PLAN or SIMULATOR; do not invent LIVE FABRIC evidence.

DE15-13 - Permissions, connections and secrets

GUIDED | Teaching Book p42 | route page 14

Starting state

Open Teaching Book page 42. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Separate identity, connection and secret responsibilities; write one least-privilege access path and one blocked action.

Expected result

permissions, connections and secrets with one explicit validation and an evidence-kind label.

Why this check matters

The action is accepted only when object/state, identity or grain, expected vs actual control, and failure/recovery meaning are explicit.

Save before moving

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DE15-14 - Monitoring, reliability and cost awareness

GUIDED | Teaching Book p45 | route page 15

Starting state

Open Teaching Book page 45. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Define run status, duration, row/freshness controls, alert condition, recovery action and cost/capacity signal.

Expected result

monitoring/reliability/cost runbook with one explicit validation and an evidence-kind label.

Why this check matters

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Save before moving

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DE15-15 - Git, deployment and environment separation

GUIDED | Teaching Book p48 | route page 16

Starting state

Open Teaching Book page 48. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Map Git branch/review -> dev -> test -> prod; separate paths/connections/IDs; define post-deploy validation and rollback.

Expected result

Git/deployment/environment separation with one explicit validation and an evidence-kind label.

Why this check matters

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Save before moving

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DE15-16 - End-to-end Fabric data engineering case

GUIDED | Teaching Book p51 | route page 17

Starting state

Open Teaching Book page 51. Keep the protected Review closed. Use the normal Retail Orders fixture when the lesson references data.

Follow with me once

Build source -> Bronze -> Silver -> Gold -> SQL consumer plus identity, run IDs/counts/quarantine/version/monitor evidence. Label LIVE FABRIC vs OFFLINE PLAN vs SIMULATOR.

Expected result

end-to-end Fabric evidence index with one explicit validation and an evidence-kind label.

Why this check matters

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